

Frank Nielsen, Sony CSL. Selected publications (2023-2026)

Journals

- Equivariant log-extrinsic statistics on symmetric cones, *Information Geometry* 2026
- Generalized Legendre Transforms Have Roots in Information Geometry, *Entropy* 2026
- Mirror Descent and Exponentiated Gradient Algorithms Using Trace-Form Entropies, *Entropy* 2025
- A geometric modeling of Occam's razor in deep learning, *Information Geometry* 2025
- Two types of geometric Jensen–Shannon divergences, *Entropy* 2025
- Symplectic Bregman divergences, *Entropy* 2024
- Fast Proxy Centers for the Jeffreys Centroid: The Jeffreys–Fisher–Rao Center and the Gauss–Bregman Inductive Center, *Entropy* 2024
- Information measures and geometry of the hyperbolic exponential families of Poincaré and hyperboloid distributions, *Information Geometry* 2024
- Variational representations of annealing paths: Bregman information under monotonic embedding, *Information Geometry* 2024
- On the f-divergences between densities of a multivariate location or scale family, *Statistics and Computing* 2024
- Divergences Induced by the Cumulant and Partition Functions of Exponential Families and Their Deformations Induced by Comparative Convexity, *Entropy* 2024
- What is... an inductive mean?, *Notices of the American Mathematical Society* 2023
- A simple approximation method for the Fisher–Rao distance between multivariate normal distributions, *Entropy* 2023
- On f-Divergences Between Cauchy Distributions, *IEEE Transactions Information Theory* 2023

Conferences

- Geometric Jensen-Shannon divergence between Gaussian measures on Hilbert space, *GSI* 2025
- Curved representational Bregman divergences and their applications, *GSI* 2025
- A Rate-Distortion View of Uncertainty Quantification, *ICML* 2024
- Hyperbolic Embeddings of Supervised Models, *NeurIPS* 2024
- Optimal transport with tempered exponential measures, *AAAI* 2024
- Quasi-arithmetic Centers, Quasi-arithmetic Mixtures, and the Jensen-Shannon-Divergences, *GSI* 2023
- Simplifying momentum-based positive-definite submanifold optimization with

applications to deep learning, ICML 2023

- Partial K-Means with M Outliers: Mathematical Programs and Complexity Results, International Conference on Optimization and Learning 2023

Book chapter

Approximation and bounding techniques for the Fisher-Rao distances, Handbook of Statistics 2024

Workshops

- Hilbert geometry of the symmetric positive-definite bicone, NeurIPS Workshop on Symmetry and Geometry in Neural Representations, NeurReps 2025
- A New Geometric Approach of Adaptive Neighborhood Selection for Classification, NeurIPS Workshop on Symmetry and Geometry in Neural Representations , NeurReps 2024
- Fisher-Rao and pullback Hilbert cone distances on the multivariate Gaussian manifold with applications to simplification and quantization of mixtures, 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG), 2023
- Non-linear Embeddings in Hilbert Simplex Geometry, 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG), 2023

Technical reports

- Dual-connection midpoint matrix means and their Gauss composition, arXiv:2609.07551
- Minimum enclosing Bregman balls made easy, arXiv:2607.24197
- Group invariance of f-divergences and the Fisher-Rao distance, arXiv:2606.25790
- Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv:2606.09077
- Generalising maximum mean discrepancy: kernelised functional Bregman divergences, arXiv:2604.24047
- Quadratic polarity and polar Fenchel-Young divergences from the canonical Legendre polarity, arXiv:2603.04812
- Geometric structures and deviations on James' symmetric positive-definite matrix bicone domain, arXiv:2603.02483
- Scales of Fréchet means and Karcher quasi-arithmetic means, arXiv:2511.21173
- Hilbert geometry of the symmetric positive-definite bicone: Application to the geometry of the extended Gaussian family, arXiv:2508.14369
- pyBregMan: A Python library for Bregman Manifolds, arXiv:2408.04175
- Tempered Calculus for ML: Application to Hyperbolic Model Embedding, arXiv:2402.04163